

ARINDAM SHARMA

MS Automotive and Motorsports | Vehicle Dynamics and Control

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Professional Profile

Masters student in Automotive Engineering & Motorsports at Politecnico di Milano specialising in vehicle dynamics and dynamic system simulation, with four years of Formula Student engineering experience. Experienced in MATLAB/Simulink and Python for vehicle modelling, tyre modelling, telemetry analysis, and data-driven vehicle performance optimisation. Hands-on experience in suspension kinematics, drivetrain design, vehicle setup, on-track testing, and correlation of simulation with test data across Formula Bharat, FS East, and FS Germany. Developed longitudinal, lateral, ride, and multibody vehicle models, with a focus on translating simulation and test data into vehicle performance improvements. Utilises LLM-assisted development to accelerate technical workflows, debugging, and exploration of simulation methodologies. Former Team Captain of Pravega Racing, leading a 30-member engineering team through vehicle development and competition.

Education

MS in Automotive & Motorsports | Politecnico di Milano

Sept 2025- Ongoing

Grade : 24.78 /30

Relevant Coursework - Vehicle Dynamics, Advanced Dynamics of Mechanical Systems, Data Analysis, Control of Mechanical Systems

B.Tech in Mechanical Engineering | VIT Vellore

July 2021- July 2025

Grade : 9.20/10

Relevant Coursework - Kinematics of Machines, Fluid Mechanics, Machine Design

Experience

Team Captain | Formula Student

Pravega Racing

Dec 2023-Dec 2024

- Led a 30-member engineering team, coordinating cross-subsystem vehicle integration and engineering deliverables.
- Directed on-track testing and suspension setup optimisation for Formula Bharat 2025 using structured **data-driven tuning, telemetry analysis, and driver feedback correlation**.
- Achieved **5 kg driveline mass reduction** and **3 hp drivetrain performance gain** through sprocket geometry and differential mounting redesign, validated against baseline simulation.
- Authored the **Structural Equivalency Spreadsheet (SES)** and **Engineering Design Report** with meticulous accuracy and led Engineering Design and Cost Report events, contributing to 1st Overall in both at Formula Bharat 2025.

Drivetrain & Performance Engineer | Formula Student

Pravega Racing

Apr 2022-Jan 2025

- Contributed to scrutineering, vehicle setup, on-track testing, and real-time data analysis.
- Developed **MATLAB/Simulink gear-ratio optimisation** algorithms for longitudinal acceleration, achieving a time of **4.9s** for the Acceleration event later validated against track runs
- Executed bench testing of the powertrain along with **differential tuning** through ramp angles and preload to maximise traction.
- Contributed across multiple engineering domains during the manufacturing and assembly of two FSAE cars, supporting drivetrain, suspension, steering, electrical, and vehicle integration activities.

Systems Engineering Intern

Larsen and Toubro-PES

Jan 2025-Apr 2025

- Developed full CAD assemblies in Siemens NX for precision mechanical systems and performed analytical load calculations and ANSYS FEA to validate designs against operational load cases.
- Automated load and structural calculations using Python, improving analysis efficiency and reducing repetitive manual calculations; worked to defence-grade tolerance and documentation standards.

Key Projects

<https://github.com/arindam-30/Automotive>

Vehicle Dynamics Simulation Models

2026

- Developed **Pacejka MF5.2 tyre models** in MATLAB using TTC Hoosier tyre data, fitting pure and combined-slip characteristics across multiple vertical-load conditions.
- Built a **3-DOF longitudinal vehicle dynamics model** in MATLAB/Simulink incorporating vehicle body motion, wheel rotational dynamics, drivetrain inertia, and Pacejka-based tire forces.
- Developed a **4-wheel dual-track lateral dynamics model** from first principles to analyse yaw rate, body sideslip, and lateral acceleration under **open-loop steering manoeuvres**, accounting for nonlinear tyre forces and load-transfer effects.
- Implemented and tuned a **PI cruise controller** in MATLAB/Simulink for stable speed tracking under varying road gradients and vehicle load conditions and a torque vectoring control in the lateral model to study its effects on vehicle performance.
- Developed a **2-DOF quarter-car ride model** to evaluate suspension dynamics, dynamic wheel loads, and ride response using frequency-domain analysis and parameter sensitivity studies.
- Optimised double-wishbone pushrod suspension kinematics by analysing camber gain, roll-centre height, anti-dive, and anti-squat characteristics against target handling requirements in MSC Adams.

Data Analysis and Filtering

2025

- Processed IMU and GPS telemetry in MATLAB using **Butterworth low-pass filtering** and **signal conditioning** to extract reliable vehicle dynamics data from track tests.
- Extracted vehicle speed, steering corrections, peak lateral and longitudinal acceleration, braking performance, and transient response metrics to evaluate vehicle behaviour across test runs.
- Reconstructed **vehicle trajectories from GPS data** and identified key events such as peak-speed locations, steering corrections, braking zones, and acceleration phases.
- Developed MATLAB scripts to **automate telemetry filtering, metric extraction, trajectory visualisation**, and run-to-run comparison for repeatable vehicle performance analysis.

AI-assisted Track Grip Prediction Model

2025

- Developed a **TensorFlow deep-learning regression model** in Python to estimate track grip coefficient from vehicle-dynamics sensor data for setup optimisation.
- Engineered and normalised tyre and vehicle-dynamics datasets for training, validation, and testing of **grip-prediction models**.
- Optimised neural-network architectures and hyperparameters using MAE and RMSE to improve grip-estimation accuracy.
- Leveraged LLM/AI-assisted development for model architecture design, debugging, and technical documentation.

Stability Analysis and Control Design

2025

- Modelled dynamic systems in MATLAB using transfer functions and state-space representations to assess stability.
- Designed and tuned PI/PD controllers to improve closed-loop transient and steady-state response.
- Assessed system stability using Nyquist, Root Locus, pole, and eigenvalue analysis.
- Characterised closed-loop performance through step response, steady-state error, DC gain, and bandwidth.
- Modelled PI-based electromechanical speed control incorporating motor and mechanical dynamics.

FSAE EV Powertrain Design

2024

- Designed the rear driveline architecture for a single-motor FSAE EV, covering shaft geometry, bearing selection, differential integration, and packaging constraints.
- Developed a parameterised MATLAB/Simulink **battery model** for cell-level discharge, SOC estimation, and thermal response under FSAE acceleration and endurance duty cycles.
- Performed battery thermal-management simulations to compare air and liquid cooling architectures and determine cooling requirements.
- Developed a parameterised vehicle simulation framework to streamline design iterations and enable correlation with measured on-car test data.

14-DOF Multibody Vehicle Model in Simulink

2026-Ongoing

- Developing a **parametric double-wishbone suspension model** in Simscape Multibody for suspension geometry optimisation and kinematic analysis.
- Integrating the suspension multibody assembly into a full-vehicle model for system-level vehicle dynamics simulation and virtual testing.
- Explored the use of **MATLAB tools** such as **Virtual Vehicle Composer** for vehicle modelling to take a reference and create Reduced-Order Models for vehicle performance analysis.
- Architecting an extensible vehicle simulation framework targeting a low-cost Driver-in-the-Loop (DiL) simulator with real-time driver interaction, hardware-in-the-loop integration, and scenario-based testing.

Technical Skills

- *MATLAB*
- *Simulink*
- *Python*
- *Model-Based Design*
- *CAD- Siemens NX , Solidworks*
- *Ansys - Mechanical , Fluent*
- *CFD*
- *OpenFOAM*
- *MSC Adams*
- *VIGrade*
- *Data Analysis*

Achievements

- 3rd Overall – Formula Bharat 2025.*
- 1st Overall – Engineering Design & Cost Report, Formula Bharat 2025.*
- 1st – Business Plan Presentation, Formula Bharat 2023.*
- 2nd – Business Plan Presentation, Formula Student East 2023.*
- Top 10 – All Static Events, Formula Student Germany 2023.*
- Merit Scholarship – Year 1, VIT Vellore (top academic rank in cohort).*